THIAGARAJAR COLLEGE OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

User Manual, Formula, project overview

Title -Evaluation of Rebar-Induced Disturbances in Ultrasonic Pulse Velocity Measurements of Reinforced Concrete

T = Transit Time in micro seconds

Vm = Measured Velocity in km/sec

Vs = velocity in steel = 5.2 km/sec

L = Path length in mm (Distance between transducer and Receiver)

Ls = length of steel in mm

a = Offset distance in mm

Vc = Corrected Velocity in km/sec

K= Correction Factor

Formula:

$$v_c = k\, v_m \qquad (3)$$

where

v_m is the measured apparent pulse velocity (L/T) (in km/s);

k is the correction factor given by

$$\gamma + 2\left(\frac{a}{L}\right)\sqrt{(1-\gamma^2)}$$

in which

$$\gamma = \frac{v_c}{v_s}$$

The pulse velocity in the concrete v_c (in km/s) is given by

$$v_c = \frac{2\, av,}{\sqrt{(4a^2 + (Tv, - L)^2)}} \qquad (2)$$

provided that $v_s > v_c$

Parallel to the Rebar

Parallel to the Rebar

$$k = 1 - \frac{L_s}{L}(1-Y) \qquad (4)$$

where

L_s is the length of the bar (in mm).

Perpendicular to the Rebar

Axis of Reinforcing Bar Perpendicular to Direction of Propagation

The influence of the presence of the reinforcing bars can be calculated assuming that the pulse traverses the full diameter of each bar during its path. If there are n different bars of diameter Q_i (i = 1 to n) directly in the path of the pulse, with their axes perpendicular to the path of propagation (see Figure 8.10A), then[22]

$$\frac{V_c}{V} = 1 - \frac{L_s}{L}\left(1 - \frac{V_c}{V_s}\right) \tag{8.6}$$

where

Perpendicular to the Rebar

References

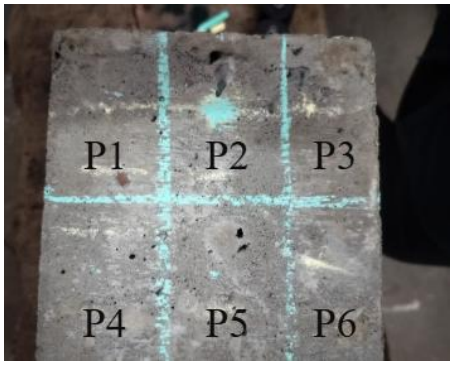
IS 10262:2019: *Concrete mix proportioning—Guidelines.* Bureau of Indian Standards, New Delhi (2019).

BS 1881-203:1986: *Testing concrete: Recommendations for measurement of velocity of ultrasonic pulses in concrete.* British Standards Institution, London (1986).

IS 13311:1992: *Non-destructive testing of concrete—Methods of test.* Bureau of Indian Standards, New Delhi (1992).

Malhotra, V.M., Carino, N.J.: *Handbook on Non-destructive Testing of Concrete.* 2nd edn. **CRC Press, Boca Raton (2004).**

Pictures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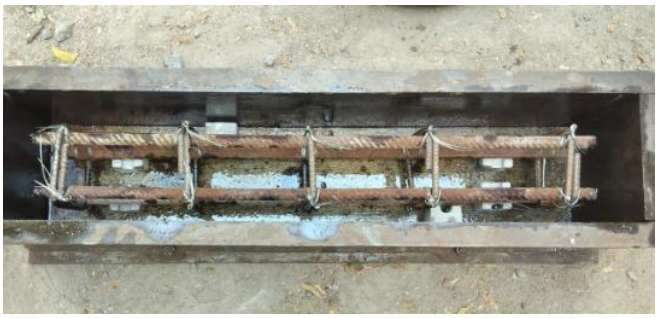
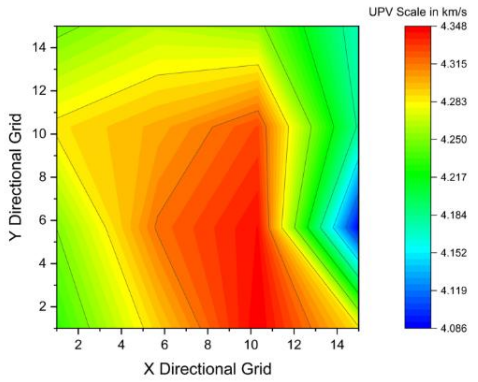
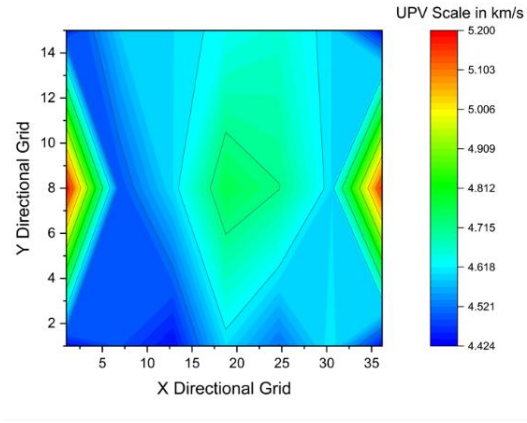
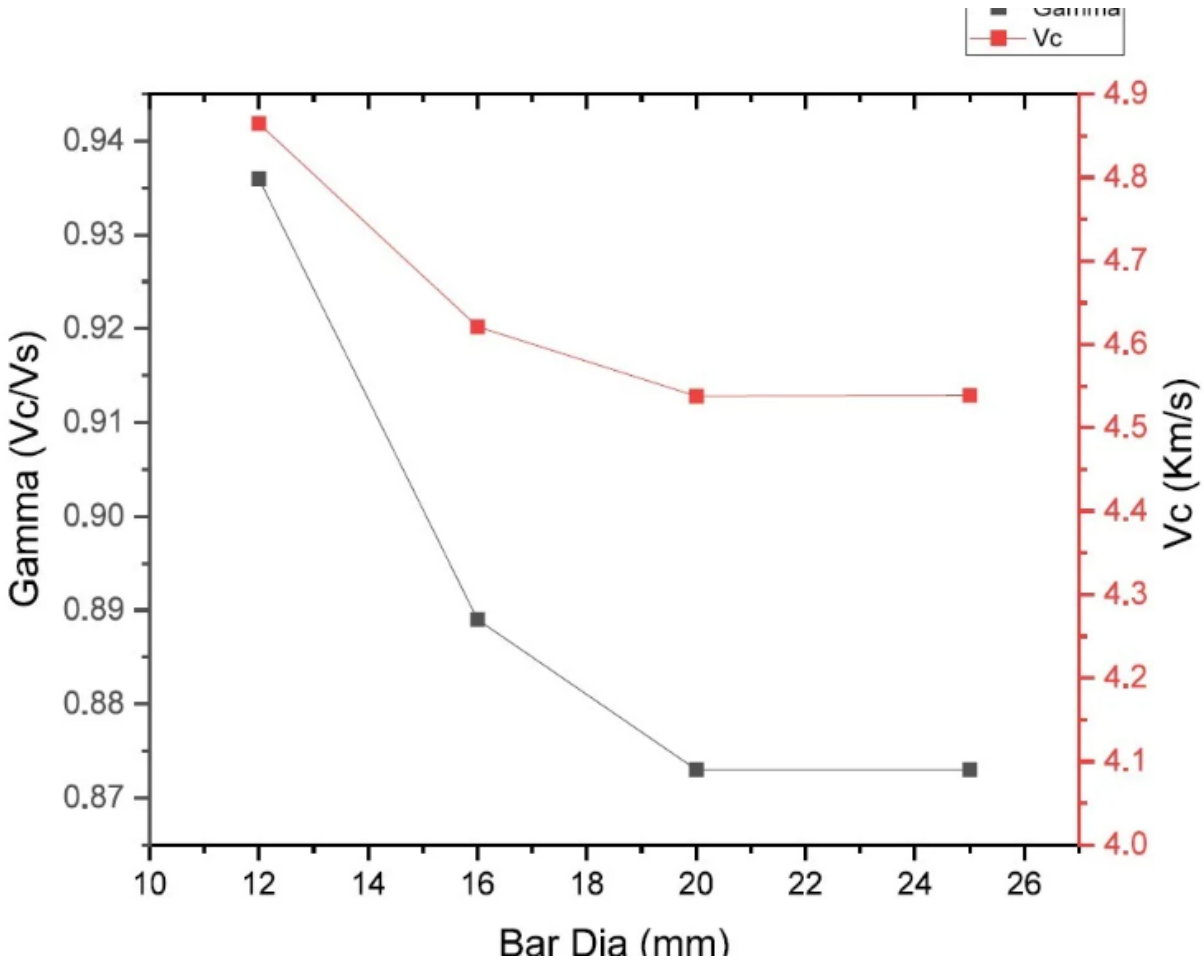






Perpendicular **Parallel**



Perpendicular to the Rebar

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INTRODUCTION
AIM
METHODOLOGY